



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 • STANDARD 6.DS.6C

Describe Data by Mean Absolute Deviation

Lesson 2-9 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find the mean absolute deviation (MAD) to describe how spread out data is.

Language Objective: I can explain my work using the words mean absolute deviation, deviation, absolute value, and spread.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mean Absolute Deviation

Deviation

Absolute Value

Spread

Data distribution

Variability



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The average distance of each value from the mean is the

2

How far a single value is from the mean is its .

3

The distance of a number from zero, always positive, is its

4

How far apart the data values are from each other is the

5

The way data values are spread out is the .

6

How much the data values differ is the .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

Both keepers allow the same average goals. Which should the coach pick and why?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Mean Absolute Deviation** — The average distance of data values from the mean; it describes spread.
Desviación media absoluta — La distancia promedio de los valores de datos respecto a la media; describe la dispersión.

☐ **Deviation** — How far a number is from the mean.
Desviación — Qué tan lejos está un número de la media.

☐ **Absolute Value** — How far a number is from zero. It is never negative.
Valor absoluto — Qué tan lejos está un número de cero. Nunca es negativo.

☐ **Spread** — How far apart the numbers are.
Dispersión — Qué tan separados están los números.

☐ **Data distribution** — How the data looks: where it sits and how spread out it is.
Distribución de datos — Cómo se ven los datos: dónde están y qué tan separados están.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Both keepers have a mean of _____. Keeper A's MAD is _____, which means _____. Keeper B's MAD is _____, which means _____. The coach should pick _____ because _____.

Di tu respuesta primero: Mi respuesta es _____ porque _____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Two data sets can share a mean but differ in how spread out they are.

Evidence: Students choose Player A as more consistent and recognize that equal means hide different spreads.

Reasoning: This evidence connects the definition of mean absolute deviation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Both keepers have a mean of ____.** Keeper A's MAD is ____, which means ____.
- **Keeper B's MAD is ____, which means ____.** The coach should pick ____ because ____.
- **My answer is ____.**

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
- **I know because ____.**

Mi afirmación es ____.

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean Absolute Deviation
2. **Notes 2:** Deviation
3. **Notes 3:** Absolute Value
4. **Notes 4:** Spread
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Mean Absolute Deviation is adding the signed deviations (value – mean) instead of their absolute values, so the positives and negatives cancel out to 0. For 6, 8, 10, 12 (mean = 9), the deviations are –3, –1, 1, 3 — they sum to 0, but that is NOT the MAD. Before averaging, take the absolute value of every deviation so each becomes a positive distance: $|-3| = 3$, $|-1| = 1$, $|1| = 1$, $|3| = 3$. Sum = 8, so $MAD = 8 \div 4 = 2$. A MAD of exactly 0 is a warning sign — it only happens if every value in the data set equals the mean.*
8. **Try It 1:** B. 2 — *Add the distances: $3 + 1 + 1 + 3 = 8$. Divide by the 4 games: $8 \div 4 = 2$. The MAD is 2 points.*
9. **Try It 2:** A. Shooter A, because a smaller MAD means scores stay closer to the average — *A smaller MAD means the scores hug the mean. Shooter A's MAD of 2 is smaller than Shooter B's 6, so Shooter A is more consistent.*